

Chapter 2 - Combustion & Thermochemistry

Relations

$$PV = N R_u T \quad | \quad N = \# \text{ Moles}$$

$$Y_i = \frac{m_i}{m_{tot}} \quad X_i = \frac{N_i}{N_{tot}}$$

$$X_i \frac{MW_i}{MW_{mix}} \quad Y_i \frac{MW_{mix}}{MW_i}$$

$$MW_{mix} = \sum_i X_i MW_i = \frac{1}{\sum_i (Y_i / MW_i)}$$

$$P_i = X_i P$$

$$\Phi = \frac{(F/A)_{act}}{(F/A)_{stoich}} = \frac{(m_{fuel}/m_{air})_{act}}{(m_{fuel}/m_{air})_{stoich}}$$

$$\% \text{ stoich air} = \frac{100\%}{\Phi} \quad \% \text{ excess air} = \frac{(1-\Phi)}{\Phi}$$

Enthalpy

$$h_{mix} = \sum_i Y_i h_i$$

$$\bar{h}_{mix} = \sum_i X_i \bar{h}_i$$

$$h_{sg}(T, P) = h_{vap}(T, P) - h_{liq}(T, P)$$

$$\bar{h}_i(T) = \bar{h}_{f,i}^\circ(T) + \Delta \bar{h}_{s,i}(T)$$

$$H_{react} = \sum_{react} N_i \bar{h}_i \quad H_{prod} = \sum_{prod} N_i \bar{h}_i$$

$$\Delta h_r \equiv h_{prod} - h_{react} \equiv -\Delta h_c \quad \text{enthalpy of react.} \quad @ T = 298^{\circ}K \quad \text{heat of combustion}$$

$$HHV \Rightarrow \text{Water Vapor Condensed (Liquid)} \quad LHV \Rightarrow \text{Water as Vapor}$$

Chapter 3

$$\text{Fick's Law} \quad \dot{n}_A'' = Y_A (\dot{n}_A'' + \dot{n}_B'') - \rho D_{AB} \frac{dx}{dx}$$

$$\rho D_{AB} \propto T^{1/2} P^0 = T^{1/2}$$

$$D_{AB} \propto T^{3/2} P^{-1}$$

$$D_{AB} = \text{Diffusivity}$$

$$\frac{\dot{Q}}{A} = \dot{Q}'' \quad \dot{n}_A'' = \frac{\dot{m}_i}{A} = \text{mass flux}$$

Units

$$R_u = 8.315 \text{ kJ/kmol}^\circ K$$

$$R = \frac{R_u}{MW}$$

Gibb's Free Energy

$$G = H - TS$$

$$\bar{g}_{i,T} = \bar{g}_{i,T}^\circ + R_u T \ln(P_i/P_0)$$

$$\Delta \bar{G}_T = \left(\sum_{\text{Products}} \bar{g}_{i,T} + \dots - \sum_{\text{Reactants}} \bar{g}_{i,T} - \dots \right) - RT \ln(K_p)$$

$$\text{Where } K_p = \frac{(P/P_0)^e (P/P_0)^f}{(P/P_0)^g (P/P_0)^h} \quad \text{or } K_p = e^{(-\Delta \bar{G}_T / R_u T)}$$

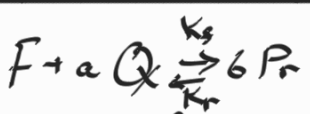
$$P_i = \text{Partial Pressure} \quad P_0 = \text{s.s. 1 atm}$$

For chemical form



Chapter 4 Chemical Kinetics

Global reaction Mechanism



$$[X_i] = \frac{X_i P}{R_u T} = \frac{P_i}{R_u T}$$

$$\frac{d[X_F]}{dt} = -k_b(T) [X_F]^n [X_O]^m \quad \text{Units: } \text{kmol/m}^3$$

Arrhenius Form

$$K(T) = A e^{(-E_a / R_u T)} \quad \text{but use } K(T) = A T^b e^{(-E_a / R_u T)} \quad \text{using Table 4.1}$$

$$A = \text{Pre exponential factor}$$

$$E_a = \text{Activation Energy}$$

For Bimolecular Rx

$$K_c = K_p$$

$$\text{IS: } 0 = -k_s [I] \dots + k_r [I]$$

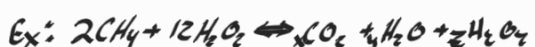
$$K_c(T) = \frac{k_s(T)}{k_r(T)} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$K_p = K_c \left(\frac{R_u T}{P_0} \right)^{c+d-a-b}$$

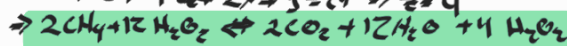
Note: $W_i = \frac{d[X_i]}{dt}$ $\rightarrow$ concentration $\rightarrow$ Reaction Rate

$$\dot{n}_i''' = W_i MW_i \left[\frac{kg}{m^3 s} \right]$$

$$\rightarrow \text{mass generation Rate/unit Volume}$$



	R	P
C	2	x=2
H	32	2y+2z=32 $\Rightarrow y=16-z \Rightarrow y=12$
O	24	2x+2y+z=24 $\Rightarrow z=4$



$$X_{H_2O} = \frac{16}{20} = 0.8$$

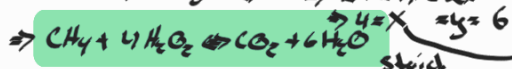
Considering 3 moles CH_4 and $\Phi = 1.5$

$$\Phi = \frac{(F/A)_{act}}{(F/A)_{stoich}} = 1.5 \Rightarrow F/A = (F/A)_{stoich} 1.5$$

$$n = \frac{F}{1.5 (F/A)_{stoich}} = \frac{3}{1.5} = 2 \text{ moles } H_2O_2$$

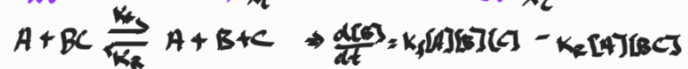


	R	P
C	1	1
H	4+2x	2y+4+2x $\Rightarrow y=2+x$
O	2x	2+y+2x $\Rightarrow 2+2+x=2x \Rightarrow 4=x \Rightarrow y=6$



For fast radical and Partial Equilibrium $\Rightarrow \frac{d[X_i]}{dt} = 0$

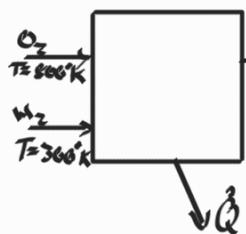
$$\frac{d[X_i]}{dt} = \sum (\text{Rate of production of } X_i) - \sum (\text{Rate of destruction of } X_i)$$



* Note is $\Delta \bar{G}_r$ is (-) and $\ln(K_p)$ is (+) > 1 reactants are favored is $\Delta \bar{G}_r$ is (+) and $\ln(K_p)$ is (-) Products are favored.

$$D^2(t) = D_0^2 - Kt \quad \text{D}^2 \text{ Law}$$

$$\text{or } \frac{dD^2(t)}{dt} = K \quad K = \text{evap. Const.}$$



Recall: 1st Law

$$\dot{Q} - \dot{W} = \dot{m} \Delta h = \dot{m} (h_p - h_r)$$

$T_{ref} = 0^\circ K$
 $P = \text{Satm}$

given: $\bar{h}_{f, H_2}(0) = \bar{h}_{f, O_2}(0) = 0 \text{ kJ/kmol}$

$\bar{h}_{f, H_2O}(0) = -238,600 \text{ kJ/kmol}$

$\bar{h}_{f, OH}(0) = -38,600 \text{ kJ/kmol}$

$\frac{\dot{Q}}{\dot{m}} = h_p - h_r = 187 \text{ kJ/kg}$

$\Phi = 0.5$

$MW_{O_2} = 32 \text{ kg/kmol}$
 $MW_{H_2} = 2 \text{ kg/kmol}$
 $\bar{c}_{p, H_2} = 40 \text{ kJ/kmol K}$

Stoich: $H_2 + \frac{1}{2} O_2 \rightarrow H_2O$

Since $\Phi = 0.5$



$\Rightarrow \dot{m}_{O_2} = \frac{\sum \dot{m}_i MW_{O_2}}{\sum \dot{m}_i} = \frac{(2 \times 16) + (1 \times 32)}{3} = 22.68 \text{ kg/kmol}$

$y_{O_2} = x_{O_2} \frac{MW_{O_2}}{MW_p} = \frac{1}{3} \frac{32}{22.68} = 0.470$ *Outlet stream mass fractions*

$y_{H_2O} = 1 - y_{O_2} = 0.530$

$h_r = \frac{\bar{h}_r}{MW_r} = \frac{x_{H_2O} \bar{h}_{f, H_2O} + x_{O_2} \bar{h}_{f, O_2}}{x_{H_2O} MW_{H_2O} + x_{O_2} MW_{O_2}}$
 $\bar{h}_{H_2} = 0 + 40(300 - 0) = 12,000$
 $\bar{h}_{O_2} = 0 + 40(800 - 0) = 32,000$

$\Rightarrow h_r = \frac{\frac{1}{2}(12,000) + \frac{1}{2}(32,000)}{\frac{1}{2}(2) + \frac{1}{2}(32)} = 1244 \text{ kJ/kg}$

$h_p = \frac{\bar{h}_p}{MW_p} = \frac{x_{H_2O} \bar{h}_{f, H_2O} + x_{O_2} \bar{h}_{f, O_2}}{22.68 \text{ kg/kmol}} = \frac{\frac{2}{3}(-238,600 + 40 T_p) + \frac{1}{3}(40 T_p)}{22.68}$

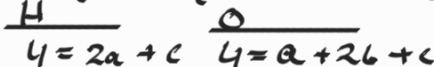
$\frac{\dot{Q}}{\dot{m}} + h_r = h_p$

$\Rightarrow -187 + 1244 = -6995.85 + 1.7637 T_p$

$\Rightarrow T_p = 4594^\circ K$ Temp of product stream

Consider,

Chemical Equilibrium to include H_2O , O_2 , OH



$\Rightarrow a = 2b$

for elementary: $K_p = e^{\frac{(-\Delta G^\circ)}{R_u T}} \quad \Delta G^\circ = 2\bar{g}_{OH}^\circ - \bar{g}_{H_2O}^\circ - \frac{1}{2}\bar{g}_{O_2}^\circ \quad @ T_{ref} = 0^\circ K$

$K_p = e^{\frac{(-\Delta G^\circ)}{R_u T}} = \frac{(P/P_0)^{c_{OH}}}{(P/P_0)^{2c_{H_2O}} (P/P_0)^{c_{O_2}}} = \frac{x_{OH}^c}{x_{H_2O}^2 x_{O_2}^c} \left(\frac{P}{P_0}\right)^{\frac{1}{2}}$

* High pressure implies $H_2O + \frac{1}{2} O_2$ is the favored Rx

Then, $\frac{\dot{Q}}{\dot{m}} = h_p - h_r \Rightarrow -187 = a[\bar{h}_{f, H_2O} + \bar{c}_p(T_p - 0)] + b[\bar{h}_{f, O_2} + \bar{c}_p(T_p - 0)] + \bar{c}_p[\bar{h}_{f, OH} + \bar{c}_p(T_p - 0)] - 2[\bar{h}_{f, H_2} + \bar{c}_p(300 - 0)] - 2[\bar{h}_{f, O_2} + \bar{c}_p(800 - 0)]$

Solve for T_p

Dam Kohler (Da)

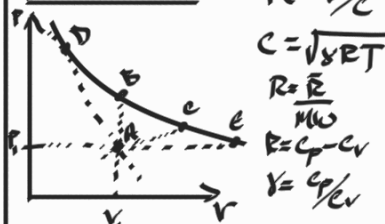
$Da \gg 1 \Rightarrow$ fast

$Da \ll 1 \Rightarrow$ slow (more unburned)

$Da = \frac{\tau_{flow}}{\tau_{chem}} \quad \tau_{flow} = \frac{L_0}{V_{rms}} \quad \tau_{chem} = \frac{S_L}{S_L}$

$V_{rms} = \sqrt{V/2}$

Detonation

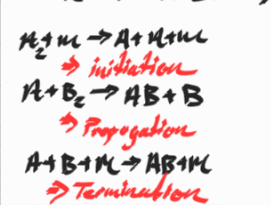


Above D: Strong weak
B-D: Strong weak
B-C: Inaccessible
C-E: weak
Below E: Strong

$-\rho_1^2 = \frac{P_2 - P_1}{v_2 - v_1} \quad v_2 = \frac{1}{\rho_2}$

$\frac{y}{y-1} (P_2 - P_1) - \frac{1}{2} (P_2 - P_1) (v_1 + v_2) - q = 0$

Chain-Chain Branching



Chapter 7

* 1-D species Continuity

$\frac{d}{dx} (\dot{m}_i Y_i) - \frac{d}{dx} (\rho D_{i,m} \frac{dY_i}{dx}) = \dot{m}_i''$

$\dot{m}_i'' = \rho Y_i (V_i - V) \quad V_i = \text{individual velocity}$
 $V = \text{Bulk flow velocity}$

* 1-D Energy Equation

$\sum (\dot{m}_i' \frac{dh_i}{dx}) + \frac{d}{dx} (k \frac{dT}{dx}) + \dot{m}_i' \frac{V_i^2}{2} = - \sum \dot{m}_i'' h_i$

* Shvab-Zeldovich Energy Equation

$\dot{m}'' \frac{d(h_{cp} T)}{dx} + \frac{d}{dx} (-\rho D \frac{d(h_{cp} T)}{dx}) = - \sum \dot{m}_i'' h_i$
Transport of enthalpy by advection
Transport of enthalpy by diffusion
heat released by reaction

* 2-D radial Conservation of Momentum

$\frac{d}{dx} (r \rho V_r V_x) + \frac{d}{dr} (r \rho V_r V_r) = \frac{d}{dr} (r \tau_{rx}) + \frac{d}{dx} (r \tau_{rx}) - r \frac{dp}{dr}$
radial momentum comes in axially
radial momentum that comes in radially
shear in r, r
shear in x -direction

gravity is often normal to radial direction

changing momentum due to pressure

Chapter 3

Fick's Law: $\dot{m}_A = \frac{1}{A} (\dot{m}_A'' + \dot{m}_A''') - \rho D_{AB} \frac{dy_A}{dx} \Big|_{x=0}$ $\rho D_{AB} \propto T^{1.5} P_0 = T^{1.5}$

Species Conservation: $\dot{m}_A''' - \nabla \dot{m}_A'' = 0$

* Mixture Fraction

$f \equiv \frac{\text{Mass of Material originating in fuel}}{\text{Mass of Mixture}}$

$$f = \frac{1 \text{ kg fuel} + \gamma \text{ kg Ox} \rightarrow (\gamma+1) \text{ kg Prod.}}{\gamma = \frac{1}{\gamma_{\text{fuel}}} + \left(\frac{1}{\gamma+1}\right) \gamma_{\text{Ox}}}$$

* Example

$C_2H_6 + a(O_2 + 3.76 N_2) \rightarrow CO + CO_2 + H_2 + H_2O + N_2 + O_2 + OH$
mole fractions: $x_{CO} = 949 \times 10^{-6}$ $x_{H_2} = 0.1488$
 $x_{CO_2} = 0.0989$ $x_{O_2} = 0.0185$
 $x_{H_2O} = 315 \times 10^{-6}$ $x_{OH} = 1350 \times 10^{-6}$

Assume balance of mixture O_2

$$f = \frac{1}{MW_{\text{mix}}} [x_{CO} MW_C + x_{CO_2} MW_C + x_{H_2O} (2 MW_H) + x_{H_2} (2 MW_H) + x_{OH} MW_H]$$

$$MW_{\text{mix}} = \sum x_i MW_i$$

$$x_{H_2} = 1 - \sum x_{i(\text{others listed})}$$

$$\rightarrow = \sum (x_i / MW_i)^{-1}$$

$$f = 0.0533$$

for $f_{\text{stoich}} = \gamma_f = \frac{MW_f x_f}{MW_{\text{mix}} x_{\text{mix}}} = \frac{MW_{\text{fuel}}}{MW_{\text{mix}} \gamma_{\text{Ox}}}$

Determining Φ from Mixture Fraction

$$\Phi = \frac{(F/A)}{(F/A)_{\text{stoich}}} \quad (F/A) = \frac{s}{1-s} \quad \text{Balance Rx for stoich}$$

$$\Phi = \frac{s/1-s}{(F/A)_{\text{stoich}}}$$

* Note: for exam most likely in $Y_i = \frac{x_i MW_i}{MW_{\text{mix}}} = \frac{m_i}{m_{\text{tot}}}$

and probably will have oxide constant in fuel will have to trace only fuel originating elements components

$$f = \frac{\text{Mass of material in fuel stream}}{\text{Mass of mixture}} = \sum \frac{MW_{\text{fuel in } i}}{MW_{\text{Ox in } i}} Y_i$$

* Example for $C_2H_6 + a(O_2 + 3.76 N_2) \rightarrow CO + CO_2 + H_2 + H_2O + O_2 + OH + N_2$

$$f_C = \left(\frac{MW_C}{MW_{CO}} Y_{CO} + \frac{MW_C}{MW_{CO_2}} Y_{CO_2} + \frac{MW_C}{MW_{CH_3OH}} Y_{CH_3OH} \right)$$

$$f_{\text{carbon}} = \frac{MW_{CH_3OH}}{MW_C} f_C$$

$$f = \gamma_{\text{fuel}} + \left(\sum \frac{MW_{\text{trace in } i}}{MW_i} Y_i \right)$$

Chapter 8 Laminar Premixed Flame

$$\rho_u S_L A = \rho_b V_b A = \rho_b V_b A$$

$u = \text{unburned}$
 $b = \text{burned}$
 $S_L = \text{laminar flame speed}$

* for burner

$$S_L = V_u \sin \alpha$$

$$\bar{T} = \frac{T_u + T_b}{2}$$

$$* \alpha \propto T_u \bar{T}^{0.5} P^{-1}$$

$$* \frac{\dot{m}'''}{\rho_u} \propto T_u T_b^n P^{n-1} e^{\left(\frac{-E_a}{R_u T_b}\right)}$$

$$1 \text{ kg fuel} + \gamma \text{ kg Ox} \rightarrow (\gamma+1) \text{ kg Prod.}$$

$$S_L = (-2\alpha(\gamma+1) \frac{\dot{m}'''}{\rho_u})^{1/2}$$

$$\alpha = \frac{2\sqrt{D} \alpha}{S_L}$$

$$\alpha = \sqrt{D} \sigma$$

$$S_L \propto \bar{T}^{-0.375} T_u^{-1/2} \left(\frac{-E_a}{2R_u T_b}\right) P^{1/2}$$

$$\sigma \propto \bar{T}^{-0.375} T_u^{-1/2} \left(\frac{-E_a}{2R_u T_b}\right) P^{-1/2}$$

$$\sigma = \frac{2\alpha}{S_L} = \frac{2K}{C_p \rho_u} \left| \dot{m}'' = \rho_u S_L \right|$$

$$\Rightarrow \frac{K}{C_p \rho_u} = \alpha \quad L_e = \frac{\alpha}{D}$$

* Metghalchi and Keck correlation

$$S_L = S_{L,ref} \left(\frac{T_u}{T_{u,ref}}\right)^{1.8} \left(\frac{P}{P_{ref}}\right)^{0.7} (1 - 2.1 Y_{H_2})$$

Example: 1-D, adiabatic, laminar

fuel flame stabilized @ Patm
* fuel propane C_3H_8
* stoichiometric mixture
* unburned gas Temp = T_{ref}
* no diluent

$$S_L = S_{L,ref} \left(\frac{T_u}{T_{u,ref}}\right)^{1.8} \left(\frac{P}{P_{ref}}\right)^{0.7} (1 - 2.1 Y_{H_2})$$

$$\Rightarrow S_L = S_{L,ref}$$

$$S_{L,ref} = (34.22) + (-138.65)(1 - 1.08)^2 = 33.33 \text{ cm/s}$$

$$S_L = V_u \sin(90) = V_u = 0.333 \text{ m/s}$$

Chapter 6 Reactions

$$\dot{h}_i = \dot{h}_{i,s} + \int_{T_{ref}}^T c_{p,i} dT \quad \dot{m}_i'' = \dot{\omega}_i MW_i$$

$$\frac{dT}{dt} = \left(\frac{Q}{V} \right) - \epsilon(\dot{h}_i, \dot{\omega}_i)$$

$$\dot{\omega} = \text{reaction rate (mole/l.s)} = \frac{dx_i}{dt}$$

$$V = \frac{M}{\sum (x_i MW_i)} \quad x_i = \frac{P MW_{\text{mix}} V_i}{P_u T MW_i}$$

$$\frac{dx_i}{dt} = \dot{\omega}_i - x_i \frac{dV}{V dt} \Rightarrow \frac{dx_i}{dt} = \dot{\omega}_i - x_i \left(\frac{\epsilon \dot{\omega}_i}{\epsilon x_i} + \frac{1}{T} \frac{dT}{dt} \right)$$

Constant Volume Stiff Mass Reactions

$$\frac{dT}{dt} = \left(\frac{Q}{V} \right) + R_u T \epsilon \dot{\omega}_i - \epsilon \dot{h}_i \dot{\omega}_i$$

$$\frac{dP}{dt} = R_u T \epsilon \dot{\omega}_i + R_u \sum x_i \frac{dT}{dt}$$

$$\frac{dx_i}{dt} = \dot{\omega}_i$$

Well Stirred Reactor

$$\frac{d\dot{m}_i}{dt} = \dot{m}_{i,in} - \dot{m}_{i,out} + \dot{m}_i''' V$$

$$\dot{m}_i'' = \dot{\omega}_i MW_i$$

$$0 = \dot{m}_i (Y_{i,in} - Y_{i,out}) + \dot{\omega}_i MW_i V$$

$$0 = 0 + \dot{\omega}_i + \epsilon \dot{h}_i (h_{u,in} - h_{u,out}) \Rightarrow \dot{\omega}_i = \dot{h}_i (h_{u,in} - h_{u,out})$$

$$\dot{\omega}_i = \dot{h}_i \left[\sum Y_{i,out} h_{f,i} - \sum Y_{i,in} h_{f,i} \right]$$

$$\text{from C.O. Mi } \dot{m}_{in} = \dot{m}_b \Rightarrow \rho_u V_u = \rho_b V_b \Rightarrow V_b = \frac{\rho_u}{\rho_b} V_u$$

$$\text{Assume I.G.L. } \rho_u = \frac{P}{R_u T_u}$$

$$C_3H_8 + a(O_2 + 3.76 N_2) \rightarrow 3CO_2 + 4H_2O + 3.76a N_2$$

$$a = 5$$

$$MW_{Rx} = \sum x_i MW_i = \frac{(944)}{14.5 + 3.76a} + \frac{(5)32}{14.5 + 3.76a} + \frac{3.76a(28)}{14.5 + 3.76a}$$

$$= 29.46 \text{ kg/kmol} \Rightarrow \rho_u = 1.194 \text{ kg/m}^3$$

$$P_b = \frac{P}{\frac{R_u T_b}{MW_{Rx}}} \quad T_b = T_{ad} = 2260 \text{ K from Table}$$

$$MW_{Rx} = \sum x_i MW_i = 28.32$$

$$\Rightarrow P_b = \frac{101.325}{28.32 (2260)} = 0.1524 \text{ kg/s}$$

$$\Rightarrow V_b = \frac{P_u}{P_b} V_u = 2.58 \text{ m}^3$$

Chapter 9 Laminar Diffusion Flame

$$L_s \approx \frac{3}{8\pi} \frac{Q_F}{D Y_{F, \text{stoch}}} \quad | \quad Q_F = \text{Volumetric flow of fuel or flame} \quad f_{\text{stoch}} = \frac{1}{s+1}$$

$$Y_{Pr} = 1 - Y_F - Y_{Ox}$$

$$Y_F = \frac{f - f_{\text{stoch}}}{1 - f_{\text{stoch}}} \quad (\text{w/ flame sheet approx}) \quad \Phi = \frac{s}{1-s}$$

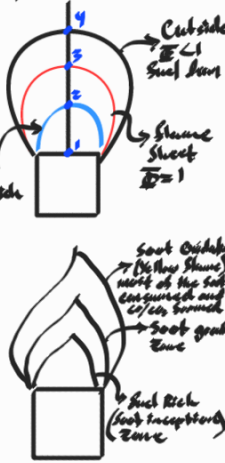
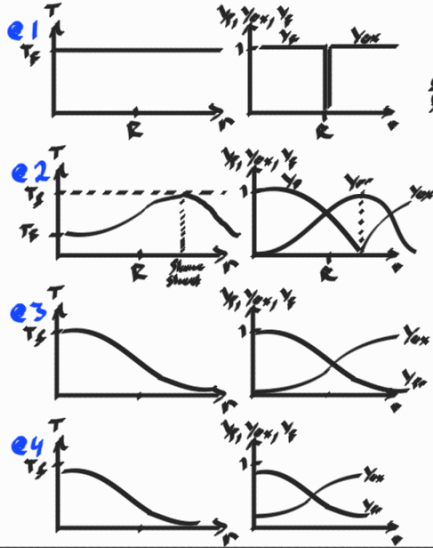
Non Reacting Jets



* Entrainment drives the rate @ which Ox can mix w/ fuel
 $\dot{m} V = \text{Constant}$ $\dot{m}_F V_F = \dot{m}_{Ox} V_{Ox}$ as $x \uparrow$ $V_F \downarrow$ $\dot{m}_{Ox} \uparrow$

$$\dot{m}_{\text{jet}}(x) = \dot{m}_{\text{fuel}}(x) + \dot{m}_{\text{Ox}}(x)$$

Constant $\dot{m}_{\text{fuel}}(x)$ must increase



* Mixture Fraction Diff. Flame

Inside Flame sheet $Y_F = \frac{s - s_{\text{stoch}}}{1 - s_{\text{stoch}}}$ $Y_{Pr} = 1 - Y_F - Y_{Ox}$

Outside Flame sheet $Y_F = 0$ $Y_{Pr} = 1 - Y_{Ox}$

@ Flame sheet $Y_F = 0$ $Y_{Pr} = 1$

Chapter 12 Turbulent Premixed Flames

$$S_T = \frac{\dot{m}}{A \rho_u} \quad | \quad \dot{m} = \rho_u A_{\text{duct}} \bar{V}_{\text{duct}}$$

$$\bar{A} = \text{Average Surface area flame}$$

$$\bar{V}_u = \frac{P}{\rho_u \bar{A}}$$

A equations

$$\bar{A} = \left(\frac{1}{2} h L \right) 4$$

$$\bar{A} = \frac{1}{2} (h_1 + h_2) 4 L$$

$$\bar{A} = \frac{1}{2} (2\pi r_1 + 2\pi r_2) L$$

$$\bar{A} = \pi S (r_1 + r_2) \quad | \quad S = \sqrt{(r_1 - r_2)^2 + L^2}$$

$\hookrightarrow$ Arc length

Radiant fraction (Rf)

$$= \frac{\text{radiation heat}}{\text{Energy content fuel (Total Energy released)}} = \frac{Q_{\text{rad}}}{\dot{m} \Delta h_c}$$

Pre-mixed Rf $\approx 10\%$

Non-Premixed Rf $\approx 30-50\%$

Chapter 13 Turbulent Non-Premixed

if buoyancy dominated, $L^* = 13.5 R_f^{0.75}$ $|$ $R_f \leq 5$

if momentum dominated, $L^* = 23$ $|$ $R_f \geq 5$

$$L^* = \frac{L_s S_s}{d_j \sqrt{\frac{\rho_c}{\rho_{ao}}}}$$

d_j = initial jet diameter

$S_s = f_{\text{stoch}}$

$$Fr_g = V_c S_s^{0.75}$$

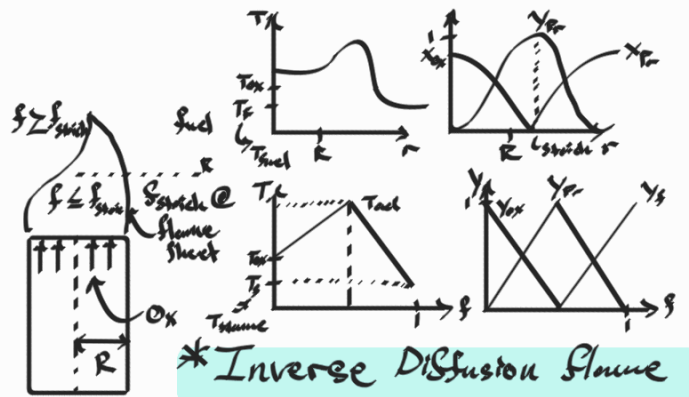
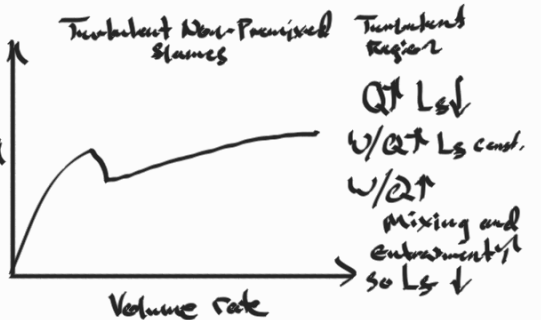
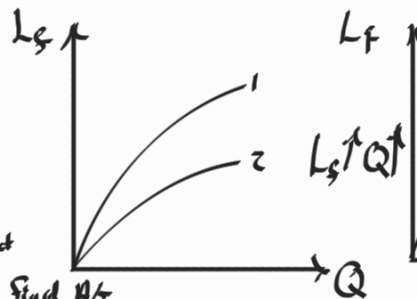
$$\left(\frac{\rho_c}{\rho_{ao}} \right)^{1/4} \left(\frac{\Delta T_s}{T_o} g d_j \right)^{1/2}$$

$\Delta T_s = T_s - T_o$

$V_c = \dot{m}_c / \rho_c A_{\text{duct}}$

$S_s = \frac{1}{(A_F) + 1}$

Find A_F from chemical balance



* Inverse Diffusion Flame

Spreading Angle α

$$\alpha = \tan^{-1} \left(\frac{L_s}{R_c} \right)$$

$R_c = \frac{\dot{m}_c}{\rho_c V_c R}$

$Q = \frac{\dot{m}_c}{\rho_c V_c}$

Find $Y_{F, \text{stoch}}$ to find $L_s = \frac{3}{8\pi} \frac{Q_F}{D Y_{F, \text{stoch}}}$

Roper's Correlation

$$L_{s, \text{theory}} = \frac{Q_F \left(\frac{T_o}{T_s} \right) \left(\frac{T_o}{T_s} \right)^{0.67}}{4\pi D \ln \left(\frac{1}{2} \right)}$$

$\left(\frac{T_o}{T_s} \right)^{0.67}$ so rather small (Rf) ratio

$$L_{s, \text{exp}} = \frac{1330 Q_F \left(\frac{T_o}{T_s} \right)}{\ln \left(\frac{1}{2} \right)}$$

$T_o = T_{\text{fuel}}$ $T_s = T_{\text{flame}}$

Froude #: $Fr = \frac{\text{momentum}}{\text{Buoyancy}} = \frac{(V_c I Y_{F, \text{stoch}})^2}{a L_s}$

for $Fr \gg 1$ momentum dominated

for $Fr \ll 1$ Buoyancy dominated

$a = 0.6 g \left(\frac{T_o}{T_s} - 1 \right)$

$V_c = \text{velocity @ exit}$

$I = \text{Normalized} = \frac{\text{momentum actual}}{\text{momentum ideal}}$

$= \frac{S_{c, \text{actual}}}{\dot{m}_F V_c}$